Example 7.17 Find the volume of the iron used to make a hollow cylinder of height 9 cm and whose internal and external radii are 21 cm and 28 cm respectively.

Solution Let r, R and h be the internal radius, external radius and height of the hollow cylinder respectively.

Given that, r =21cm, $R = 28$ cm, $h = 9$ cm

Now, volume of hollow cylinder $= \pi(R^2 - r^2)h$ cu. units

$$= \frac{22}{7}(28^2 - 21^2) \times 9$$

$$= \frac{22}{7}(784 - 441) \times 9 \ = 9702$$

Therefore, volume of iron used $= 9702$ cm^3

Example 7.18 For the cylinders A and B (Fig. 7.27),

(i) find out the cylinder whose volume is greater.

(ii) verify whether the cylinder with greater volume has greater total surface area.

(iii) find the ratios of the volumes of the cylinders A and B.

Solution

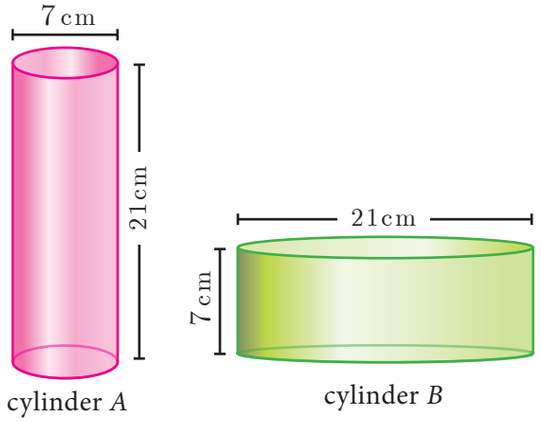
(i) Volume of cylinder $= \pi r^2 h$ cu. units

Volume of cylinder $A = \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 21$

$= 808.5$ cm^3

Volume of cylinder $B = \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times 7$

$= 2425.5$ cm^3



cylinder A cylinder B

Fig. 7.27

Therefore, volume of cylinder B is greater than volume of cylinder A.

(ii) T.S.A. of cylinder $= 2\pi r(h + r)$ sq. units

$$\text{T.S.A. of cylinder } A \ = 2 \times \frac{22}{7} \times \frac{7}{2} \times (21 + 3.5) = 539 \text{ cm}^2$$

$$\text{T.S.A. of cylinder } B \ = 2 \times \frac{22}{7} \times \frac{21}{2} \times (7 + 10.5) = 1155 \text{ cm}^2$$

Hence verified that cylinder B with greater volume has a greater surface area.

(iii) $\frac{\text{Volume of cylinder A}}{\text{Volume of cylinder B}} = \frac{808.5}{2425.5} = \frac{1}{3}$

Therefore, ratio of the volumes of cylinders A and B is 1:3.

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